

MLOps

Group Assignment

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Calendar

Mon Jan 19

Open Group Assignment

Fri Feb 6

Deadline to submit assignments

Sat Feb 7 - Sun Feb 8

Final Exam

Mon	Tue	Wed	Thu	Fri	Sat	Sun
					17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1-Feb
2	3	4	5	6	7	8

1. Telco Churn

Business Case:

We have noticed an unusual large number of customers leaving to another telco provider in the last year. We are a reference telco company in our market with more than 15 million customers. Our call center handles more than 100K daily calls from customers, mainly complaining about the invoice or the service.

Scores are used to mark potential customer in risk of churn:

- In future marketing retention campaign
- When customer interacts with the company the agent can provide an excellent service

2. Manufacturing Predictive Maintenance

Business Case:

Stopping the production line is very expensive. It would be better if we could anticipate when to stop the line for maintenance although we don't want to risk our warranty contract with the equipment provider. We have 5 factories with 4 production lines. Each line includes a centrifuge equipment as one of the main elements.

Scores are used to calculate the risk of failure and the expected time to failure of the centrifuge equipment in the line:

- The mean time to failure in this equipment is 3 months
- We are receiving monitoring data each 15 seconds

3. Auto Insurance Dynamic Pricing

Business Case:

We need to calculate the premium pricing for the customers. This is done once a year for renewals and we have created specific models for new customers

Scores are used to calculate the price of the premium that is going to be cheaper for low-risk customers:

- The pricing application is central to all the company and is used in the renewal
- All of our agents in all the branches have access to the application in order to price new customers
- We are going to use IoT data from the vehicle that we can use to model driving behavior. This is offered as an option to customers at a reduced premium

4. Demand Forecasting for Utilities

Business Case:

We need to calculate the demand forecasting of electricity for each hour in order to prevent blackouts during this heat wave

Scores are used to calculate the total amount demanded for each region:

- We have the individual metering values every 15'
- We have the historical outages
- We think it could be interesting to use weather data

5. Smart Agent for Tax Agency

Business Case:

We need to help citizens with their taxes and streamline the number of agents that we need. We are going to include a speech to text and generative AI application that helps the citizen with their questions.

Both the web page and the mobile application will have access to the smart agent:

- We have data from most frequent questions
- We have recorded more than 10.000 citizen interactions
- The Smart Agent will present best scoring answer to the citizen
- When the tax campaign begin, we are attending an average of 100K interactions each day

Deliverable

- Provide **a set of questions with answers** related to different phases in the MLOps cycle for the use case provided
- Summarize and argument **what is the biggest challenge** from MLOps point of view in the use case
- Focus on main messages
- Send pdf with 10 pages max